

AMAN SHASTRY

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EDUCATION

University of Michigan (GPA: 4.00)

Bachelor of Science in Engineering Data Science

Current Coursework: Machine Learning, Web Systems, Applied Regression Analysis

Past Coursework: Data Structures and Algorithms, Statistics and AI, Applied Linear Algebra, Multivariable Calculus

Ann Arbor, MI

December 2027

SKILLS

Programming languages: Python, Dart, C++, React

Software: NumPy, PyTorch, Pandas, Flutter, Git, Linux Scripting

Hardware: Microcontroller programming (Arduino), sensor and IoT integration, circuitry

WORK EXPERIENCE

ExaThought

Software Developer Intern

Remote

September 2023 – Dec 2023

- Developed and deployed Password Reset and Contact Us flows for an e-signature application using NextJS framework
- Explored creating API endpoints using NodeJS for integration into developed pages
- Exposed to the agile development flow through daily meetings and bi-weekly sprints and code version control through Git

PROJECT EXPERIENCE

Multi-Agent Framework for Medical Diagnosis

Independent Project

Remote

June 2025

- Developed an agentic medical assistant to simulate a collaborative multi-specialist diagnosis (GP, cardiologist, neurologist)
- Leveraged dynamic routing, memory, and human-in-the-loop feedback to generate patient-specific diagnoses and reports
- Integrated PubMed as a reasoning tool for Retrieval Augmented Generation

Scientific Computation Model Context Protocol Server

Independent Project

Remote

June 2025

- Deployed a Model Context Protocol (MCP) server to allow large language models to accurately perform linear algebra and vector calculus operations
- Achieved 25000+ monthly server calls with 90%+ server response success rate
- Used NumPy and SciPy libraries to build tools and resources, and used Smithery to deploy the server

Michigan Data Science Team

Project Lead

Ann Arbor, MI

January 2025 – Present

- Co-led project on detecting Alzheimer's disease from MRI scans of brains, teaching team members about modern Machine Learning Techniques: CNNs, Residual Networks finetuning, Feature Visualization, and Model Ensemble techniques
- Prepared visual explanations and code notebooks for application-focused teaching of modeling techniques and best use cases
- Previously worked on breast cancer detection through diverse approaches including CNNs, Google's Vision Transformer, and ResNet, for both binary cancer detection and multi-class subtype classification
- Achieved 99% accuracy with Google's Vision Transformer and 60% accuracy with ResNet

Google Student Developer Club

President

Ann Arbor, MI

September 2024 – Present

- Partnered with HomeRise to lead an 8-person team in building a modular web application for tracking staff members and residents in a scalable and secure manner; provided constant and transparent progress updates to client and delivered final application in timely manner.
- Led a 10-person team to develop an end-to-end Audio Journal application including a custom fine-tuned WavLM model for sentiment analysis achieving 60% classification accuracy; assessed team members' strengths to assign specialized tasks, adapted responsibilities as needs evolved, and maintained transparency through regular updates and cross-functional collaboration
- Designed and developed a modular web application for the Humane Society of Huron Valley to optimize volunteer and foster pet management, automating and digitising manual workflows and reducing administrative overhead by an estimated 30-40%
- Partnered with Nevada Homelessness Alliance to connect users to essential resources like food, shelter, and health through a mobile application, built in partnership with Developers for Good

PROGRAMS/SOCIETIES

Eta Kappa Nu, Beta-Epsilon Chapter

Active Member

Ann Arbor, MI

December 2025